

ENDALKACHEW YAREGAL

Software Engineering Student · Backend-Leaning Full-Stack Developer

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SUMMARY

Software Engineering student and backend-leaning full-stack developer with hands-on experience designing, building, and deploying real-world web applications, SaaS platforms, AI-powered features, payment workflows, and background processing systems. Strongest in TypeScript/JavaScript, Node.js, PostgreSQL, Redis, Docker, and modern web architecture, with a particular interest in backend systems, reliability, APIs, databases, and distributed workloads. Built applications addressing real Ethiopian use cases, including Amharic language processing, Ethiopian calendar support, TeleBirr and CBE payment workflows, and low-bandwidth user experiences. Seeking a 4-month software engineering internship starting February 2027 (onsite or hybrid) to contribute to production systems while learning from experienced engineers.

EDUCATION

Woldia University — Ethiopia

Bachelor of Science in Software Engineering | In Progress

Focused on software engineering fundamentals while independently developing and deploying production-oriented applications. Practical focus on backend architecture, databases, APIs, distributed workloads, cloud deployment, and modern full-stack development.

TECHNICAL SKILLS

Languages TypeScript, JavaScript, Python, SQL

Frontend React, Next.js, Vite, Tailwind CSS, Zustand, TanStack Query, TanStack Router, Zod

Backend & Databases Node.js, Express, PostgreSQL, Prisma, Redis, BullMQ, Firebase/Firestore, JWT, Better Auth

AI & Integrations Google Gemini, Stripe, TeleBirr, CBE Payments, Nodemailer, Resend, S3-compatible Storage, Backblaze B2

Infrastructure & Tools Docker, Docker Compose, Vercel, Railway, Render, Cloudflare Workers, Git, GitHub, CI Fundamentals

SELECTED PROJECTS

A-kiyon FCMS — Fitness Center Management System | 2026 · Live

Production fitness-center management platform built for the Ethiopian market.

- Built member management, subscription management, revenue analytics, role-based access, receipt printing, and multi-currency payment workflows.
- Implemented Ethiopian calendar support and soft-delete/rollback functionality for operational workflows.
- Built image upload and optimization pipeline using Backblaze B2 and Sharp, including WebP conversion and storage monitoring.
- Implemented Firebase authentication, Firestore batch operations, CSV exports, and responsive management interfaces.

Tech: Next.js, TypeScript, Firebase, Firestore, Tailwind CSS, shadcn/ui, Recharts, Vercel | [Live](#) · [GitHub](#)

Amharic Auto-Caption Generator & CaptionForge | 2026 · Production

AI-powered Telegram and browser-based captioning tools for mixed Amharic-English audio and video.

- Built a production Telegram bot that converts mixed Amharic-English audio into SRT subtitles using Gemini Flash.
- Implemented a database-backed job queue using PostgreSQL/Supabase, SKIP LOCKED job claiming, dual workers, automatic retries, and restart resilience.
- Integrated CBE payment verification into the processing workflow.
- Built CaptionForge, a browser-native video captioning tool with waveform editing and client-side subtitle processing using WebCodecs.
- Developed a 97-test automated test suite focused on reliability and failure handling.

Tech: Python, Telegram Bot API, Gemini, Supabase, Next.js, WebCodecs, Wavesurfer.js, Railway

Full Stack Time Capsule | 2026 · Open Source

Microservices platform for encrypted digital time capsules with scheduled future delivery.

- Designed a microservices architecture with separate application and worker workloads.
- Implemented JWT authentication, S3-compatible storage, presigned URLs, scheduled background jobs, and automated email notifications.
- Built containerized local development with Docker Compose and automated database migrations.
- Implemented asynchronous processing with Redis and BullMQ for scheduled delivery and background workloads.

Tech: TypeScript, React, Express, PostgreSQL, Prisma, Redis, BullMQ, Docker, Nginx | [GitHub](#)

KeyForge — Node-Locked Licensing Server | 2026 · Live

Product licensing infrastructure for hardware-bound software activation and license management.

- Built a licensing server supporting online activation, offline verification, seat enforcement, device revocation, and administrative audit trails.
- Implemented Ed25519-signed license tokens with zero-dependency Node.js verification.
- Designed atomic seat limits and rate limiting to protect licensing operations from concurrent activation conflicts.

Tech: Next.js, TypeScript, PostgreSQL, Prisma, Vercel, Neon | [Live Demo](#)

FeedWise — AI-Powered Customer Feedback SaaS | 2025 · Open Source

Full-stack SaaS platform that collects customer feedback through QR codes and converts feedback into AI-powered business insights.

- Built a monorepo architecture with shared Zod contracts between frontend and backend.
- Developed Gemini-powered feedback analysis for sentiment, urgency, retention risk, and business insights.
- Implemented Stripe Basic/Pro subscriptions, webhooks, billing portal, rate-limited alerts, and daily digests.
- Added Redis caching and BullMQ background jobs for asynchronous processing.

Tech: TypeScript, React, Vite, Express, PostgreSQL, Prisma, Redis, BullMQ, Gemini, Stripe, Docker | [GitHub](#)

UniExam Hub — Contributor | 2025 · Live

Collaborative mobile-first exam preparation platform for Ethiopian university students.

- Contributed to a platform supporting multiple question types, practice/test modes, AI tutoring, progress tracking, administration tools, and payment verification.

Tech: TypeScript, React, Prisma, PostgreSQL, TanStack Query, Gemini, S3, Docker | [Live](#)

ADDITIONAL INFORMATION

- Backend-focused interests: APIs, databases, background jobs, queues, reliability, payments, and AI-assisted systems.
- Ethiopian product experience: Amharic/Ge'ez, Ethiopian calendar, TeleBirr, CBE payment workflows, and low-bandwidth/3G-friendly experiences.
- Open-source contributions and private production systems available on request.
- Languages: Amharic (Native), English (Professional).
- Internship availability: 4 months starting February 2027 (2019/06/01 E.C. / 08 Feb 2027 G.C.). Preferred arrangement: Onsite or Hybrid.